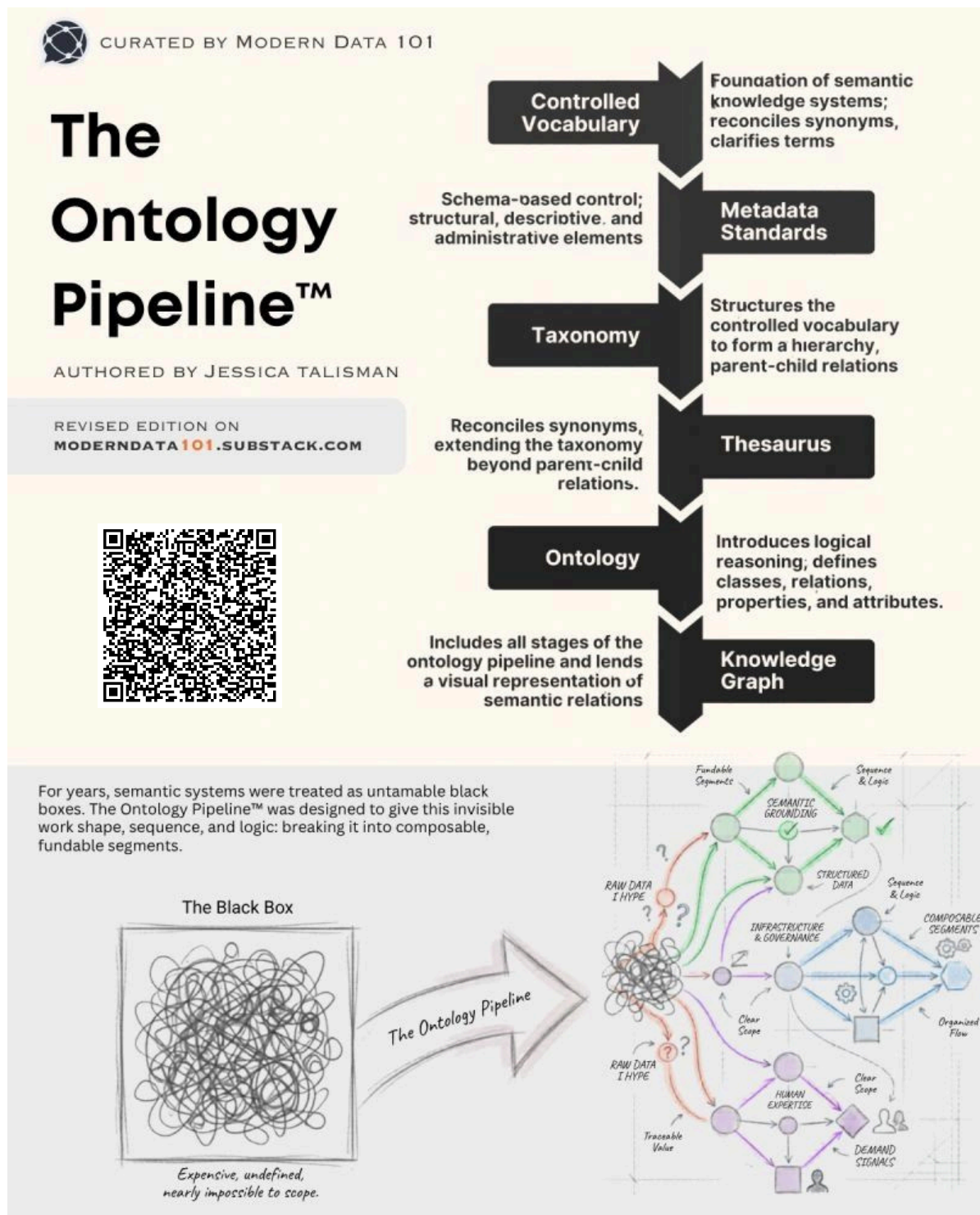




InfiniAI Pulse

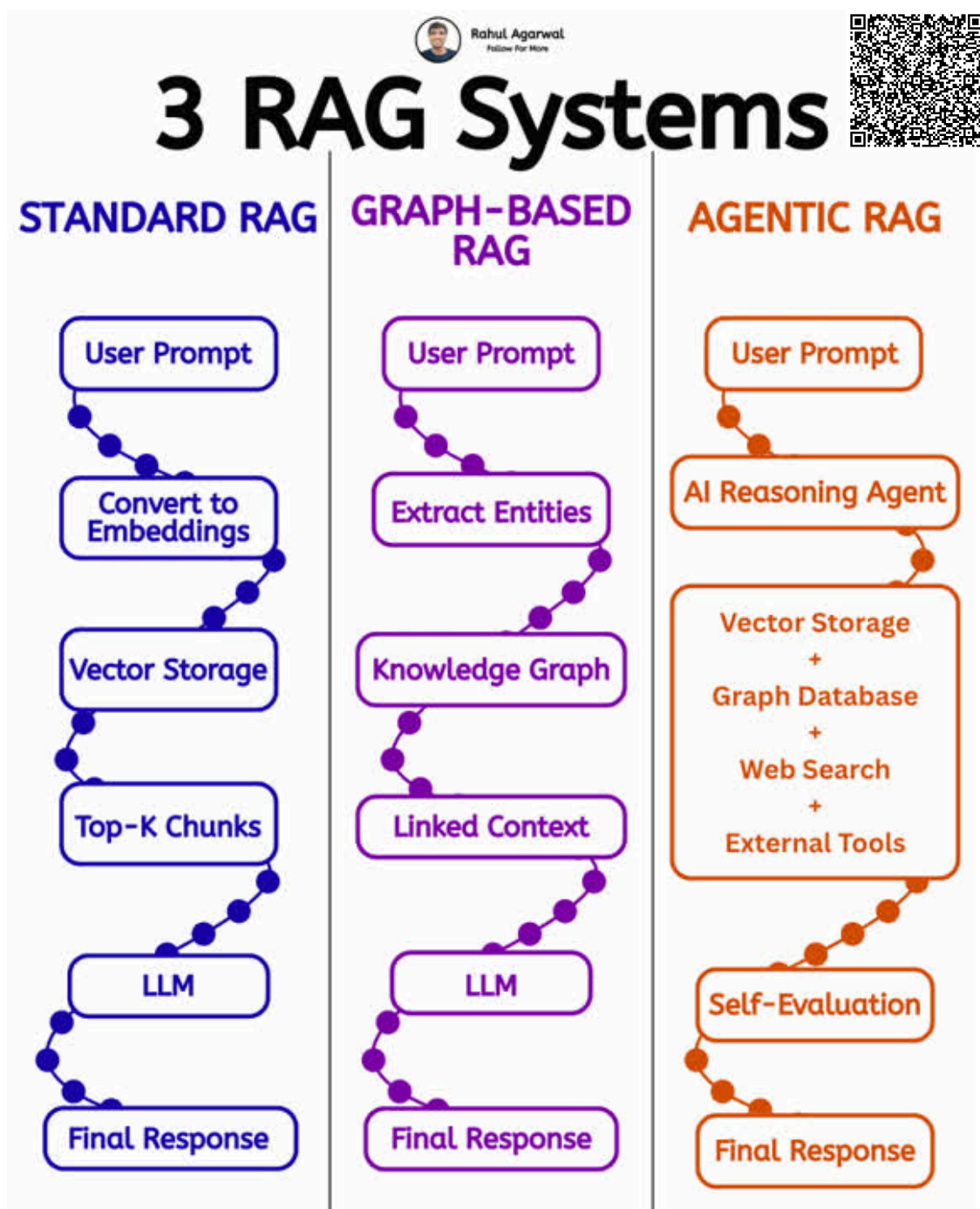
The ontology pipeline transforms raw, unstructured information into organized semantic knowledge systems. It starts with controlled vocabularies and metadata standards, then builds taxonomies, thesauri, ontologies, and finally knowledge graphs that define relationships, logic, and meaning between data entities. The goal is to move from disconnected data “black boxes” to structured, explainable, and machine-understandable knowledge architectures for intelligent AI and semantic systems.





InfiniAI Pulse

RAG systems are evolving beyond simple document retrieval. Standard RAG retrieves the most relevant text chunks from vector databases for fast responses, while Graph-based RAG improves contextual understanding by linking entities and relationships through knowledge graphs. Agentic RAG goes further by combining reasoning agents, external tools, web search, and self-evaluation to make AI systems more autonomous and adaptive. The progression is from basic retrieval → connected knowledge → intelligent autonomous decision-making.

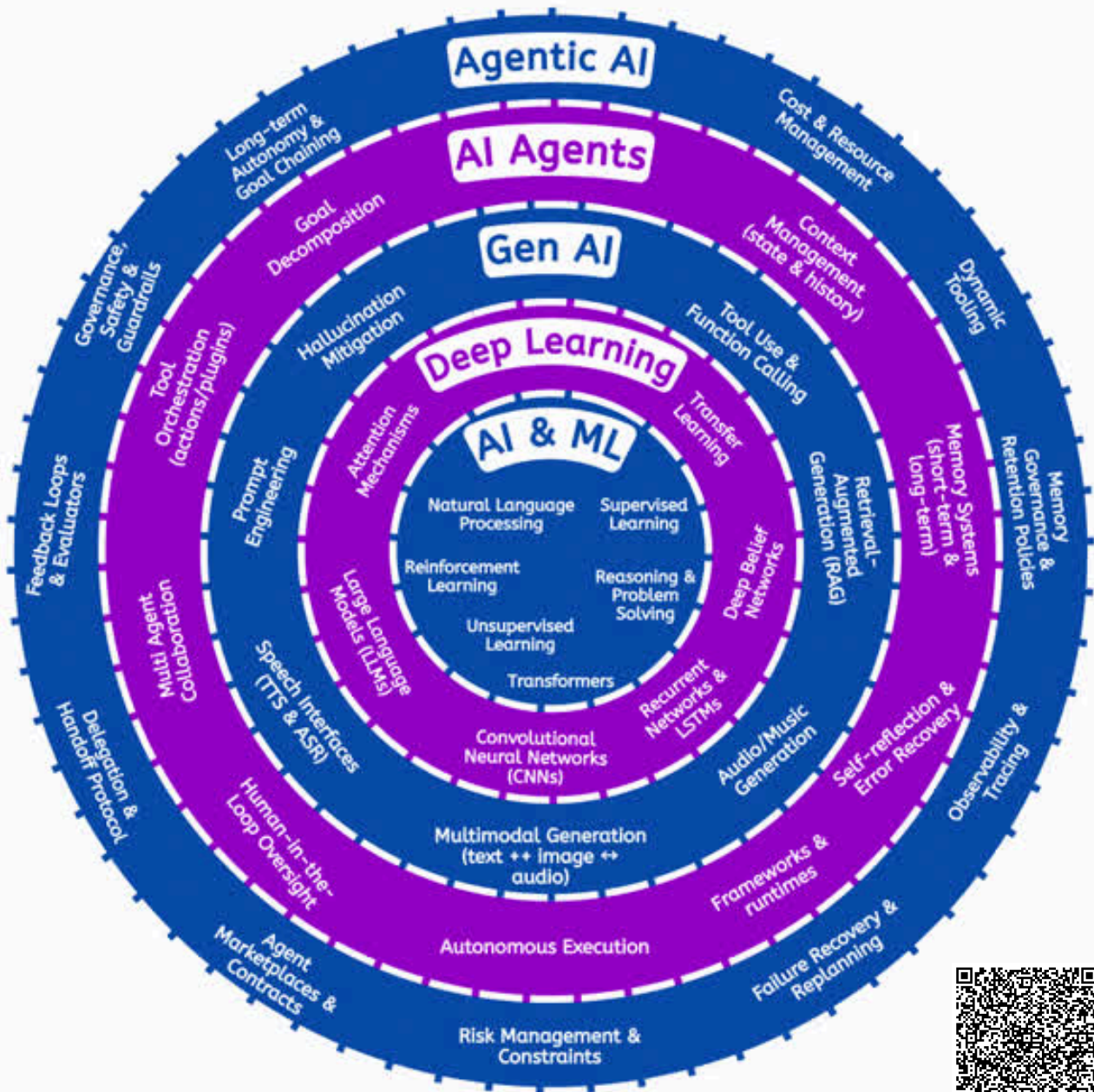




InfiniAI Pulse

AI systems are evolving in layers, from traditional AI & ML models to Deep Learning, Generative AI, AI Agents, and fully Agentic AI systems. Each layer adds new capabilities such as reasoning, multimodal generation, memory, tool usage, autonomous execution, and self-reflection. The progression shows how AI is moving from simple prediction models toward intelligent systems that can plan, collaborate, adapt, and act with increasing autonomy.

AI Systems Layers

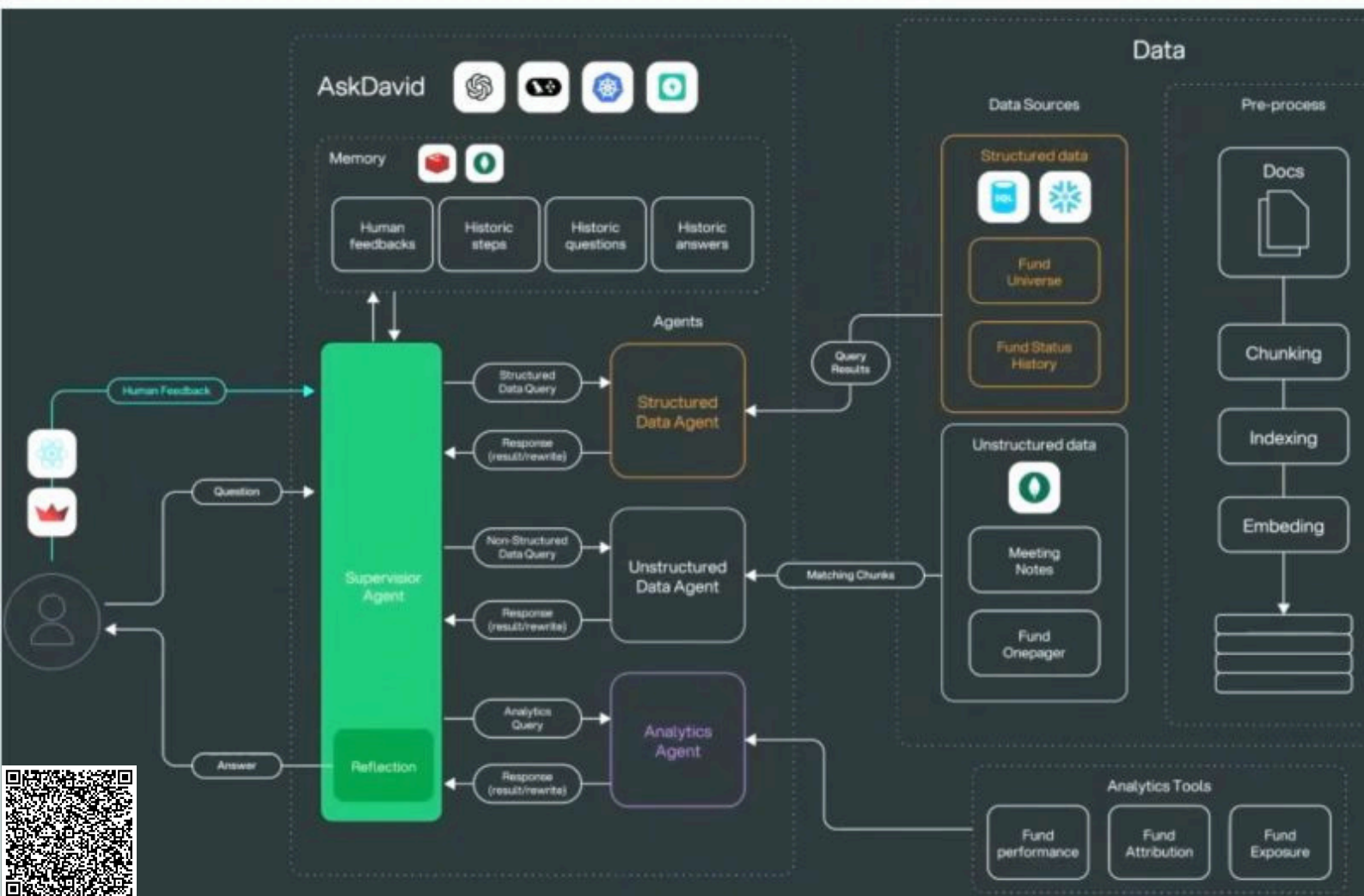




InfiniAI Pulse

Multi-agent AI systems divide complex tasks across specialized agents that collaborate through a central supervisor agent. Instead of relying on a single model, different agents handle structured data, unstructured documents, analytics, memory, and reasoning independently while sharing results dynamically. This architecture improves scalability, accuracy, context management, and autonomous decision-making for real-world enterprise AI applications.

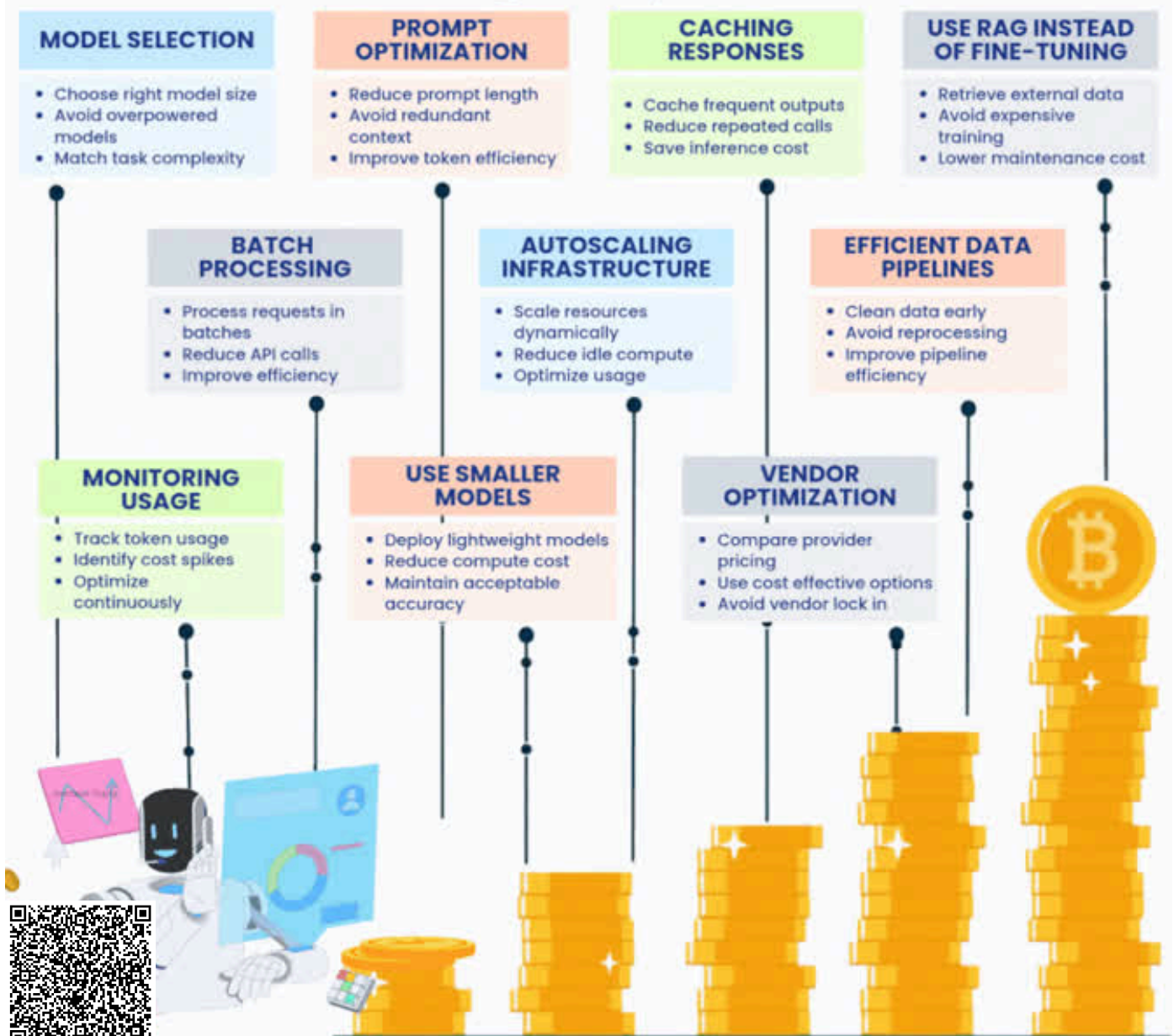
A Real Multi-Agent System



Credits: LangChain

AI cost optimization is not just about using cheaper models—it's about building efficient AI systems. Techniques like prompt optimization, caching, batch processing, autoscaling, smaller models, and RAG-based retrieval help reduce compute, API, and infrastructure costs while maintaining performance. The goal is to balance accuracy, scalability, and operational cost for sustainable production-ready AI systems.

Antrixsh Gupta
Production-Ready AI



I build real-world AI systems. DM "AI" to turn your AI idea into a production-ready system.

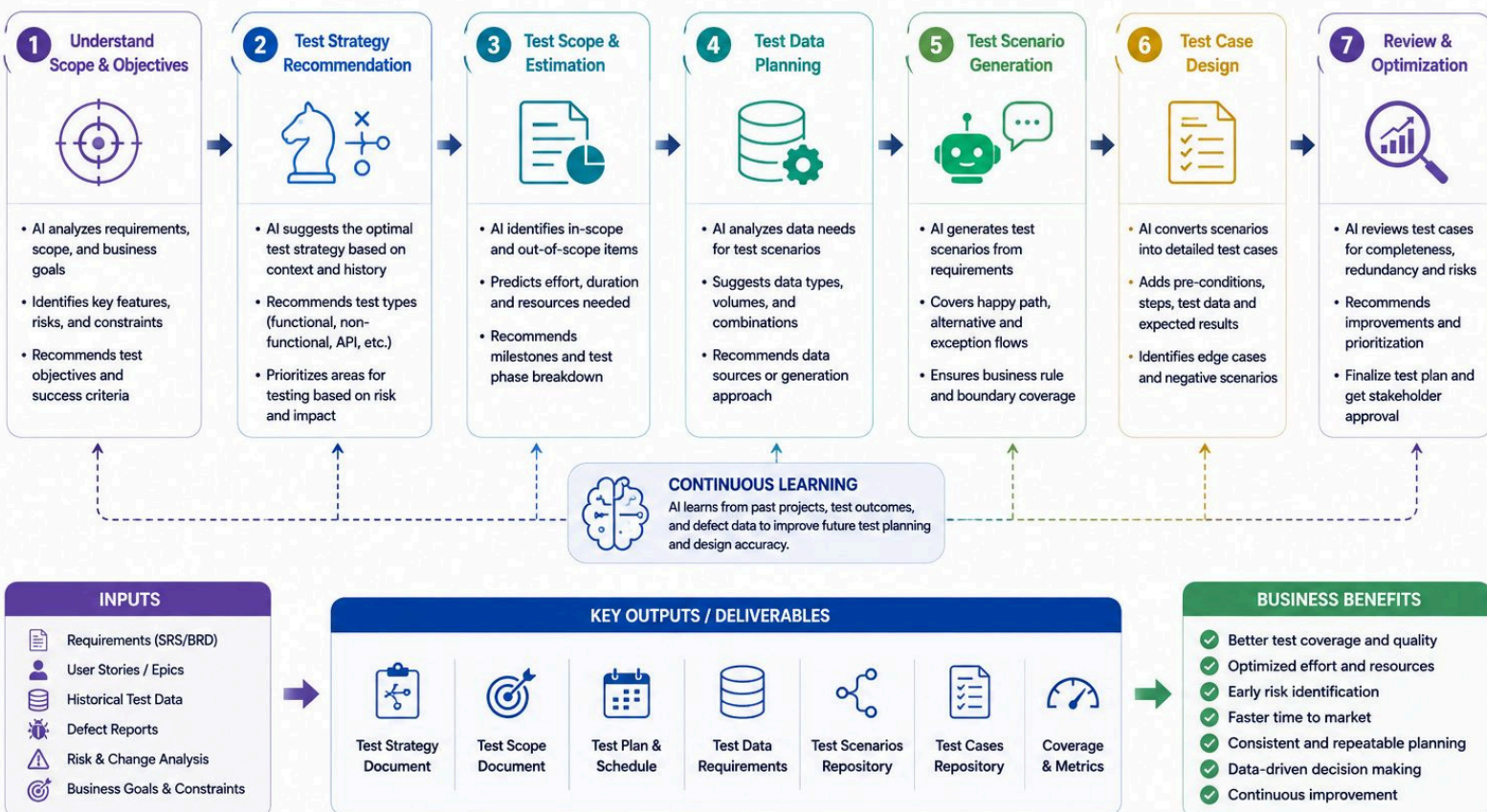


InfiniAI Pulse

AI-assisted test planning modernizes traditional software testing by automating strategy selection, scope estimation, scenario generation, test data planning, and optimization. Instead of relying only on manual effort, AI continuously analyzes requirements, risks, and historical testing data to improve coverage, accuracy, and efficiency. Rather than replacing testers, AI enhances decision-making and enables faster, smarter, and more consistent quality engineering workflows.

AI-ASSISTED TEST PLANNING & DESIGN WORKFLOW

Intelligent Planning. Smarter Design. Better Coverage.



AI doesn't replace testers. It **empowers** them to plan better, design smarter, and deliver higher quality with confidence.





InfiniAI Pulse

Generative AI is reshaping software testing by helping QA engineers automate test case generation, API testing, bug analysis, security validation, and performance testing using intelligent prompts. Instead of spending hours on manual scripting and planning, testers can use AI to improve speed, coverage, accuracy, and overall testing efficiency across the QA lifecycle.

GEN AI FOR QA

COMPLETE PROMPT CHEAT SHEET + LIBRARY

Test Cases
 API Testing
 Automation
 SQL
 JMeter
 Bug Analysis
 Security
 Bug Analysis

1 TEST CASE GENERATION

PROMPT:

Act as a Senior QA Engineer. Generate login page test cases including positive, negative, edge, usability, and security scenario. Return in priority order.

EXAMPLE OUTPUT:

- Valid login with valid credentials
- Invalid password
- Locked account
- Empty username
- SQL injection input
- Remember me validation
- ...and more

2 API TESTING

PROMPT:

Generate POST /login API scenarios. Include status codes, schema validation, auth token checks, and response time.

EXAMPLE OUTPUT:

Status Code	Scenario
200	Valid login
400	Missing field
401	Invalid credentials
429	Rate limit
✓	Token expiry validation

3 SELENIUM AUTOMATION

PROMPT:

Write Selenium Java code for Login page using explicit waits and reusable methods.

EXAMPLE CODE:

```
WebDriver driver = new ChromeDriver();
WebDriverWait wait = new WebDriverWait(
    driver, Duration.ofSeconds(10));
WebElement user = wait.until(
    ExpectedConditions.visibilityOfElement
    Located(By.id("username")));
user.sendKeys("test");
WebElement pass = driver.findElement
    (By.id("password"));
pass.sendKeys("1234");
pass.submit();
```

4 SQL VALIDATION

PROMPT:

Write SQL query to find duplicate users based on email + mobile.

EXAMPLE QUERY:

```
SELECT email, mobile, COUNT(*) as cnt
FROM users
GROUP BY email, mobile
HAVING COUNT(*) > 1;
```

5 JMETER

PROMPT:

Design JMeter plan for Login API with 1000 users, 5 min ramp-up, and SLA under 2 sec.

EXAMPLE PLAN:

- Thread Group: 1000
- Ramp-up: 300 sec
- Loop Count: 10
- Assertions: Response time < 2 sec

6 BUG ANALYSIS

PROMPT:

Analyze stack trace and suggest probable root cause, impacted module, and recommended fix.

EXAMPLE OUTPUT:

```
Root Cause: Null pointer exception
in LoginService.java line 85
Module: Authentication
Impact: Login fails for valid users
Fix: Add null check before calling
getUser() method.
```

7 SECURITY TESTING

PROMPT:

Generate OWASP based test cases for login functionality.

EXAMPLE AREAS:

- SQL Injection
- Cross Site Scripting (XSS)
- Brute Force Attack
- Session Fixation / Hijacking
- Input Validation
- ...and more

8 REGRESSION TESTING

PROMPT:

Based on checkout page changes, identify impacted regression test cases and prioritize smoke scenarios.

FOCUS AREAS:

- Payment flow
- Cart & Order summary
- Coupon / Discounts
- UI & Validation
- Notifications

9 PERFORMANCE ANALYSIS

PROMPT:

Review this JMeter report and identify bottlenecks using response time, error %, CPU, and memory metrics.

KEY METRICS:

- Response Time
- Throughput
- Error %
- CPU Utilization
- Memory Usage

10 PROMPT FORMULA (Use Everywhere)

Act as [Role] + Do [Task] + Include [Context] + Return [Format]

EXAMPLE:

Act as a Senior QA Engineer. Generate API test cases for Payment API. Include positive, negative, edge, and security scenarios. Return in table format.

TOP GEN AI TOOLS FOR QA

ChatGPT
 Claude
 Gemini
 Copilot

Faster Test Design
 Better Coverage
 Clearer Bugs
 Higher Quality

BEST PRACTICES FOR WRITING EFFECTIVE PROMPTS

- Be Specific: Use context and expectations
- Provide Details: Include inputs, constraints & examples
- Iterate: Refine prompt for better output
- Validate: Always review & validate AI outputs

SAVE & SHARE